



Smile Planet Farms



Smile Planet Farm

Through these slides, we will explore :

- Data cleaning process
- Form design logic
- Dashboard walkthrough
- Insights
- Recommendations

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PROJECT OBJECTIVES

1. Clean and structure the provided farm dataset (Egg Production 2022).
2. Design an improved farm data collection system using Google Forms.
3. Integrate cleaned + newly structured data.
4. Build an insightful, executive-ready dashboard in Looker Studio.
5. Present actionable business insights for Smile Planet Farms.



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DATA AUDIT REPORT

Dataset: Daily Egg Production & Feed Consumption

Organization: Smile Planet Farm

Period Covered: January - December, 2022

Records: 32 rows (including header-level anomaly on first date)

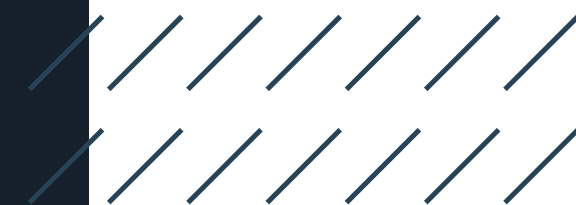
Prepared for: Dataset Familiarization

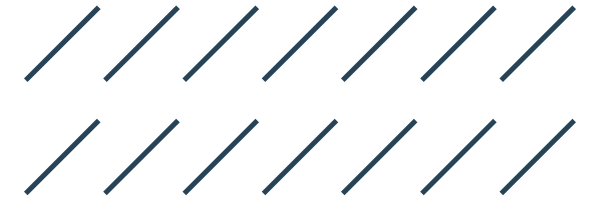
Date: March 4th, 2026



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1. EXPLORATORY DATA ANALYSIS (EDA) REPORT

1.1 Introduction to the Dataset:

This report presents a structured overview of the raw, unprocessed Egg Production Dataset 2022 obtained from Smile Planet Farms. It outlines the dataset's structure, identifies data quality issues and anomalies, and highlights key patterns observed during the initial Exploratory Data Analysis (EDA), in line with the defined business objectives.

Dataset Metadata

Data Source: Smile Planet Farms

File Format: Microsoft Excel (.xlsx)

Time Coverage: January 2022 to December 2022

Dataset Title: Egg Production Dataset 2022

Number of Sheets: 12 (monthly records)

Data Type: Tabular records capturing monthly egg production activities



b. Dataset Overview

The dataset captures daily poultry production and feed usage across two production wings of a poultry farm. It records egg output (crates and pieces), egg quality indicators (damaged and small eggs), feed consumption metrics, associated costs, and staff accountability details.

Following the audit and cleaning process, all monthly sheets (January–December 2022) were consolidated into a single **Master Dataset** to support consistent analysis, reporting, and visualization.

Key Columns

- **Date:** Daily production date
- **Week:** Week number within the month
- **Crates_Wing1 / Crates_Wing2:** Full egg crates produced per wing
- **Pieces_Wing1 / Pieces_Wing2:** Remaining egg pieces per wing
- **Total_Daily_Production:** Total eggs produced per day (in pieces)
- **Damaged Eggs:** Number of broken eggs
- **Small Eggs:** Number of undersized eggs
- **Feed Consumed in Bags:** Quantity of feed consumed daily
- **Cost of Feed Per Bag (₦):** Unit cost of feed
- **Cost of Feed Consumed (₦):** Total daily feed cost
- **Weight of Feed Bag (kg):** Total feed weight consumed
- **Picked By:** Staff responsible for egg collection
- **Approved By:** Supervisor approving the record



2. Data Quality Assessment

2.1 Missing Values

Significant missing values were observed in the raw dataset, particularly in:

- **Damaged Eggs**
- **Small Eggs**
- Staff-related fields (*Picked By, Approved By*)

2.2 Duplicate Records

- Duplicate date entries were detected (e.g., **07/01/2022**).
- One instance contained a nearly empty row alongside a fully populated record for the same date, indicating a data entry anomaly rather than a true duplicate.

2.3 Inconsistent Formats

The dataset exhibited multiple formatting inconsistencies:

- Mixed naming conventions (uppercase with spaces vs. snake_case)
- Inconsistent staff name formatting (e.g., "DAVID & SADIQ" vs "DAVID &SADIQ")
- Ambiguous date formats (DD/MM/YYYY vs MM/DD/YYYY)
- Numeric columns stored as text due to currency symbols and mixed formatting

2.4 Incorrect or Questionable Entries

- The first record for some dates contained only **Total_Daily_Production**, with all component fields missing.
- Feed consumption values appeared constant across extended periods, suggesting possible estimation rather than direct measurement.
- No validation rules initially enforced logical relationships between component fields and totals.





3. Summary of Identified Data Issues

Major Issues

- Duplicate date entries with incomplete records
- High volume of missing values in egg quality fields
- Ambiguous and inconsistent date formats
- Mixed numeric and text data types in quantitative columns
- Inconsistent staff name representations
- Lack of standard handling for blanks vs. zero values

Minor Issues

- Inconsistent column naming conventions
- Unmerged and improperly formatted week identifiers
- Lack of standardized currency and decimal precision





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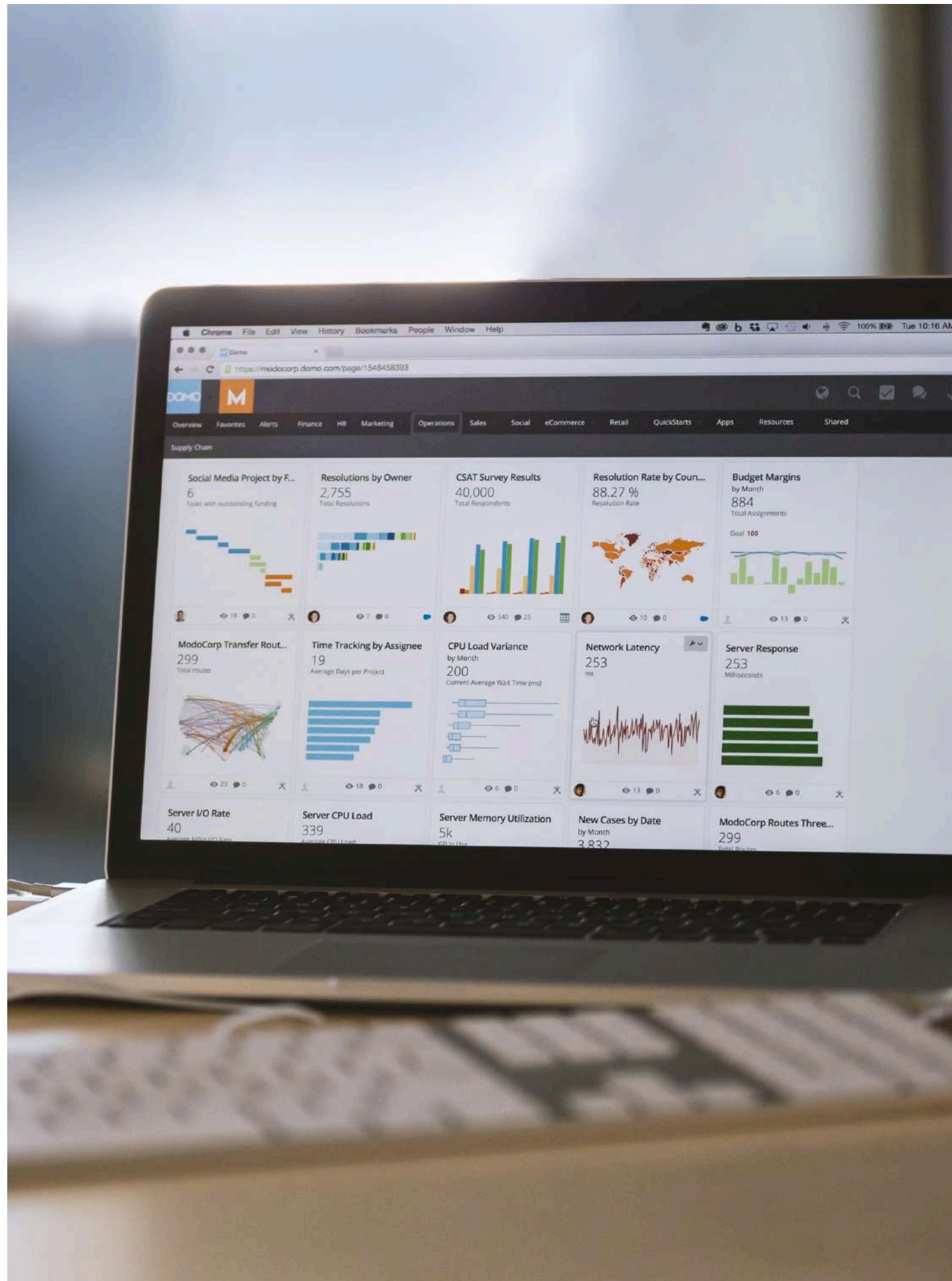
DOCUMENTATION OF GOGGLE FORM DESIGN AND DATASET RESTRUCTURING

4.1 Background and Objective:

Prior to this intervention, farm production data at Smile Planet Farms was recorded across multiple monthly Excel sheets. While this approach captured historical information, it presented several challenges, including inconsistent data entry, difficulty in validation, duplication of values, and time-consuming data cleaning before analysis.

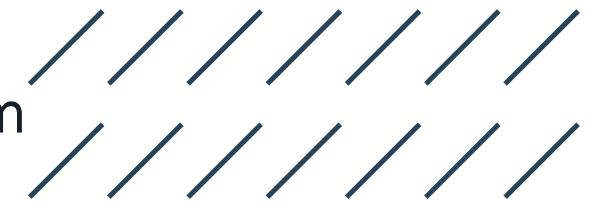
The objective of this task was therefore to:

- Design a structured Google Form to standardize daily farm data collection going forward.
- Ensure that collected data aligns with the cleaned master dataset structure used for analysis.
- Improve overall data accuracy, completeness, consistency, and usability.
- Enable seamless integration with Google Sheets for real-time data storage and analysis.



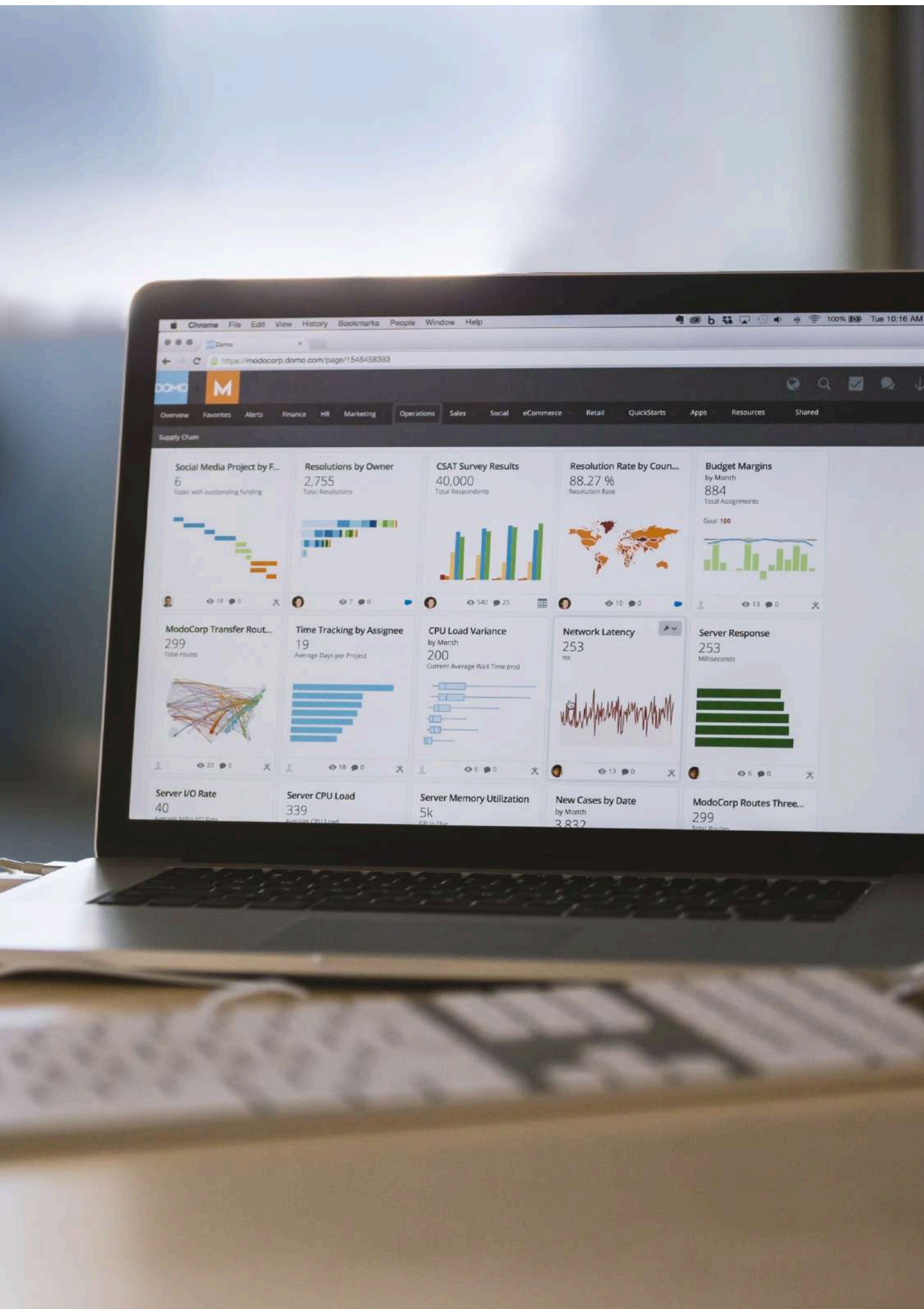


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4.2 Goggle Form Design Overview

The Google Form titled “Smile Planet Farm_Daily Egg Production & Consumption Form” was designed to capture daily operational data in a structured, validated, and analysis-ready format. The form is divided into three logical sections, each aligned with a specific data collection purpose.





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4.3 Goggle Structure and Sections

Section 1: Daily Egg Production (General Information)

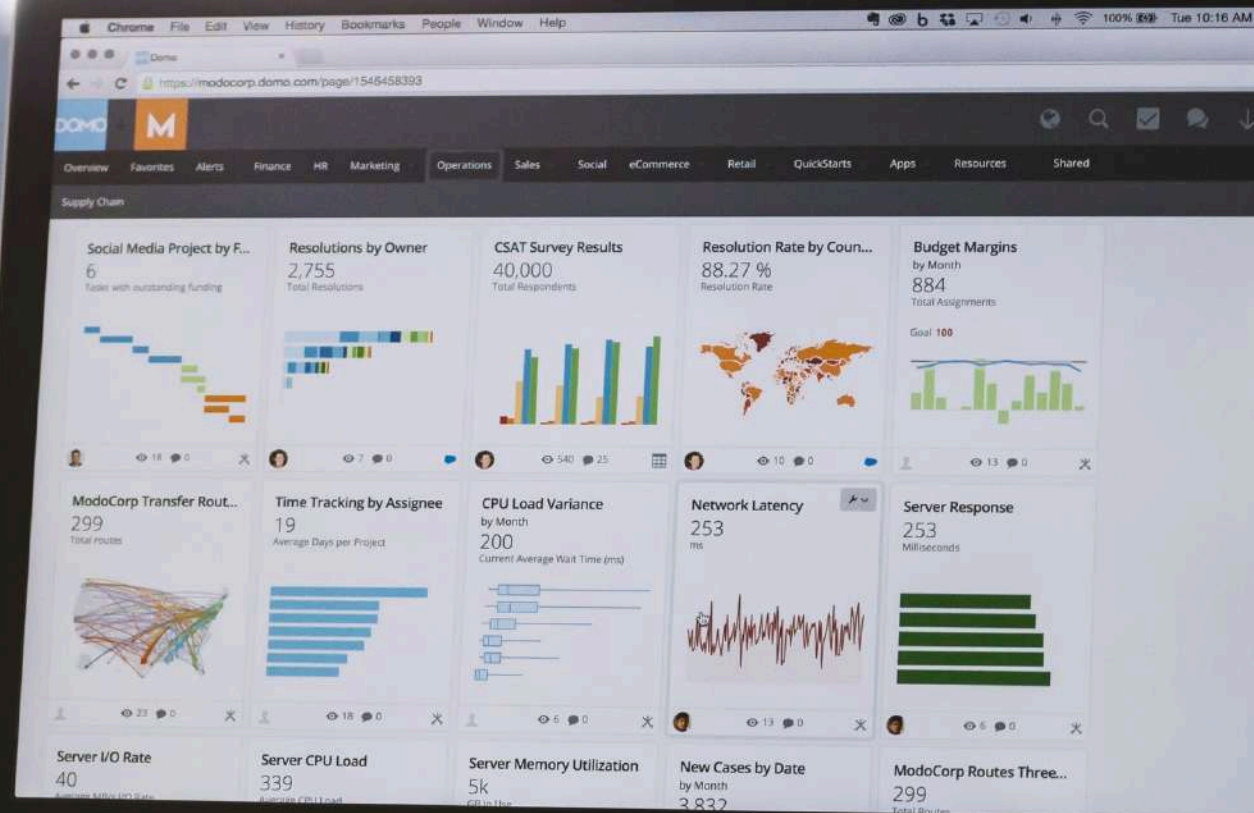
This introductory section establishes context and ensures proper classification of records.

Fields captured include:

- Email (Required): Automatically collected to track data submitters and ensure accountability.
 - Wing Selection (Dropdown – Required):Wing 1
 - Wing 2
 - Data Type Selection (Dropdown – Required):Egg Production
 - Feed Consumption

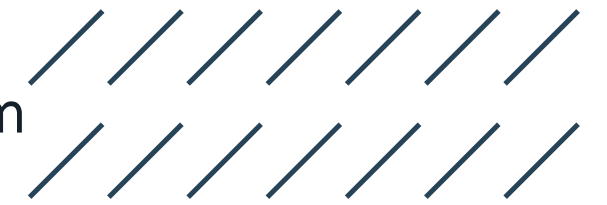
Improvement over previous structure:

- Eliminates ambiguity by explicitly separating production data from feed data.
- Enforces standardized wing identification.
- Prevents mixed or incomplete records.





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Section 2: Egg Production

This section is displayed only when “Egg Production” is selected.

Fields captured include:

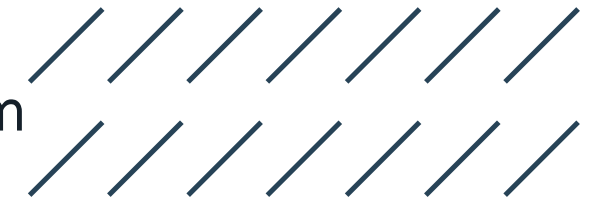
- Crates (Required – Numeric): With guidance to enter 0 when production is less than 30 eggs.
- Pieces (Required – Numeric): Restricted to values not exceeding 29.
- Damaged Eggs (Required – Numeric): Defaults to 0 when none occur.
- Small Eggs (Required – Numeric): Defaults to 0 when none occur.
- Picked By (Dropdown – Required): Predefined list of farm staff.
- Approved By (Dropdown – Required): Supervisory approval field.
- Production Notes (Optional): For operational comments or anomalies.

Improvement over previous structure:

- Replaces free-text entry with dropdowns and numeric validation.
- Ensures damaged and small eggs are always recorded, even when zero.
- Introduces traceability through “Picked By” and “Approved By”.



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Section 3: Feed Consumption

This section is displayed only when “Feed Consumption” is selected.

Fields captured include:

- Feed Consumed in Bags (Dropdown – Required): Predefined fractional values.
- Cost of Feed Per Bag (Required – Numeric Validation): Restricted to an approved price range.
- Cost of Feed Consumed (Calculated Field): Derived from quantity × unit cost.
- Weight of Feed Bag (Dropdown – Required): Standardized weight options.
- Feed Consumption Notes (Optional): For operational remarks.

Improvement over previous structure:

- Prevents unrealistic feed quantities and pricing.
- Standardizes feed measurement units.
- Reduces post-collection calculations and errors.



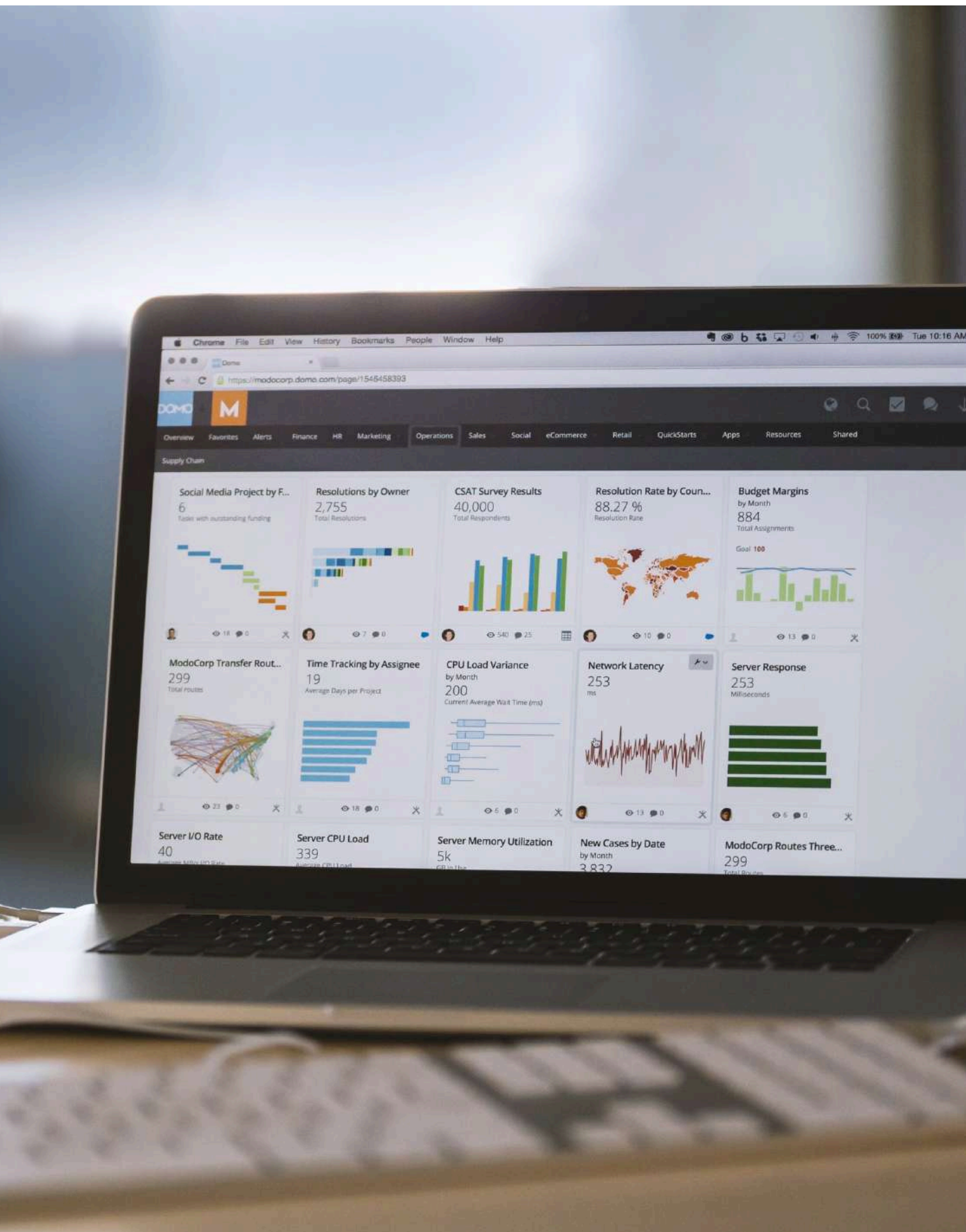
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5. DATA INTEGRATION WITH GOGGLE SHEET

The Google Form is configured to automatically submit responses into a linked Google Sheet. Each submission generates a timestamped record with standardized column headers.

This integration ensures:

- Real-time data availability.
- Elimination of manual data entry.
- Improved data timeliness and consistency.





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5.1 Challenge: Alignment with Historical Master Dataset

Although the form structure was optimized for future data collection, the existing Master Dataset had a different structure. The original dataset:

- Combined Wing 1 and Wing 2 data in single rows.
- Stored production and feed data together.
- Was organized monthly and later merged into a yearly dataset.

To ensure structural compatibility between historical data and new Google Form responses, a restructuring process was required.

5.2 Use of Google Apps Script for Dataset Restructuring

To address this challenge, a custom Google Apps Script was developed to transform the cleaned Master Dataset into a new format named: Master Dataset_(Restructured).

Purpose of the Script:

The script was designed to:

- Convert each daily record into four standardized rows:
 - Wing 1 – Egg Production
 - Wing 1 – Feed Consumption
 - Wing 2 – Egg Production
 - Wing 2 – Feed Consumption
- Match the exact column structure generated by the Google Form.
- Preserve historical data while making it analysis-ready.

5.3 Key Transformations Applied:

The script performed the following actions:

- **Timestamp Standardization:**
- Dates were safely converted into timestamps without timezone shifts.
- **Wing Separation:**
- Wing 1 and Wing 2 data were split into separate rows.
- **Data Type Classification:**
- Each row was explicitly labeled as either Egg Production or Feed Consumption.
 - **Metric Allocation:** Crates and pieces assigned only to production rows.
 - **Feed-related metrics** assigned only to feed rows.
 - **Damaged and small eggs** evenly distributed across wings where applicable.
- **Column Alignment:**
- **Output columns exactly match Google Form response headers, including:**
 - Timestamp
 - Email Address
 - Wing
 - Data Type
 - Production metrics
 - Feed metrics
 - Personnel fields

5.4 Outcome and Benefits:

The restructuring resulted in a unified dataset that:

- Seamlessly integrates historical data with future Google Form responses.
- Requires minimal cleaning before analysis.
- Supports automated dashboards and reporting.

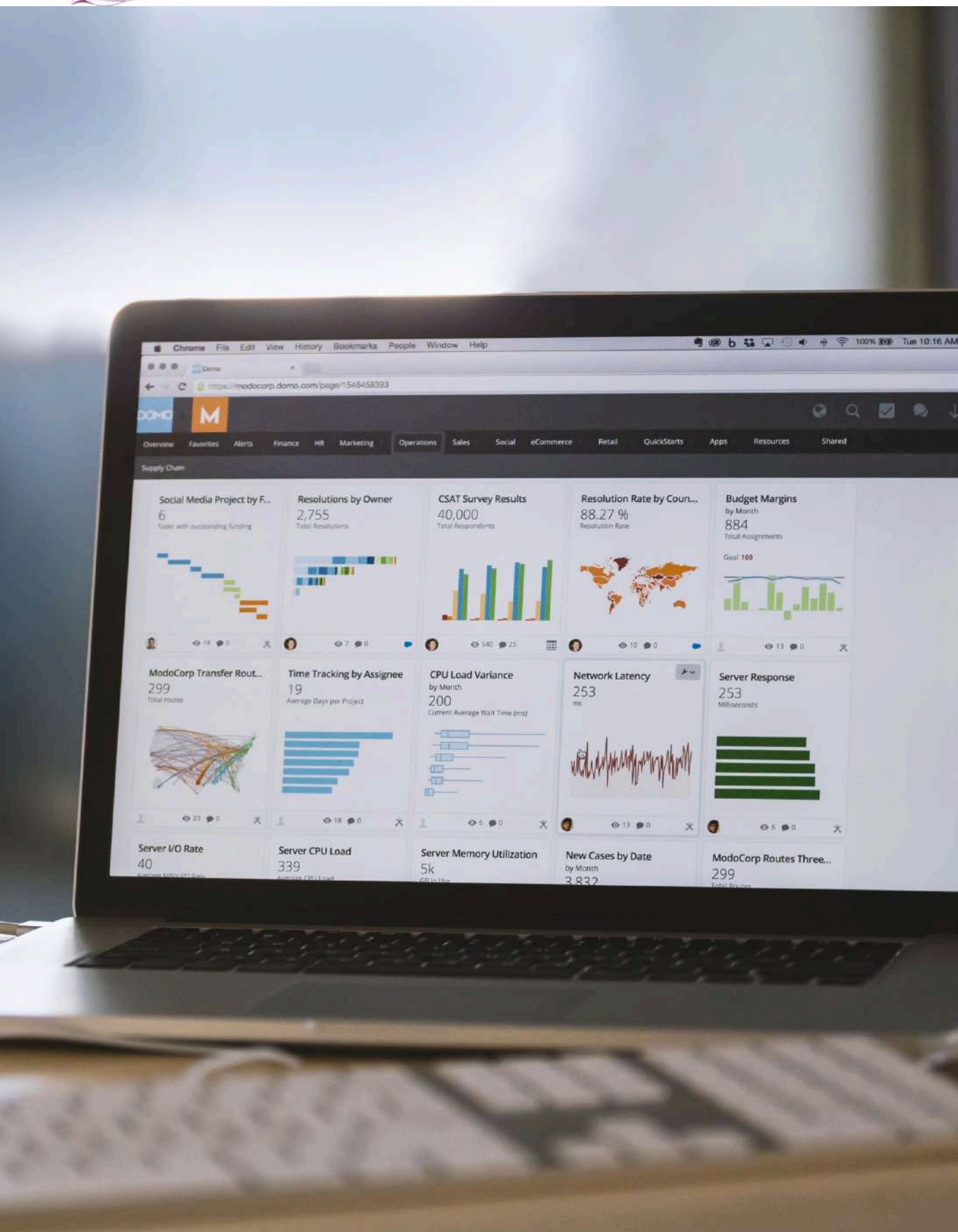


INSIGHTS AND BUSINESS ANALYSIS

6.1 Insights to the Business:

1. The calculated percentages of small eggs (0.10%) and damaged eggs (0.41%) indicated that the farm produced a very low proportion of small eggs, while the incidence of damaged eggs also remained minimal during the period under review.
2. The bar chart on the first dashboard page showed that January recorded the lowest number of egg crates produced. Production levels in the remaining months were largely consistent, with both wings maintaining nearly the same rate of output throughout the year.
3. The bar chart on the second dashboard page indicated that Wing 2 accounted for a higher share of total egg production compared to Wing 1.
4. The table displayed on the second dashboard page revealed that Shedrack recorded the highest number of damaged eggs among the staff.
5. Because wing allocation was not considered during the recording of small and damaged eggs, the bar chart on the second dashboard page showed equal counts for both wings in each case; for both small egg counts and damaged egg counts.
6. As one of the key performance indicators (KPIs), the calculated feed efficiency ratio was 181.07, indicating that approximately 181 eggs were produced for every bag of feed consumed. This implied that the consumption of 3.5 bags of feed would yield an estimated 650 eggs. However, the analysis showed that peak production periods occurred mainly in February and March, where egg output was comparatively higher relative to feed consumed. In contrast, January recorded lower egg production despite feed consumption, indicating reduced feed efficiency during that month.





6.2: Recommendations to the Business

1. Low small and damaged egg rates.

- The farm should maintain current handling, collection, and storage practices, as these have proven effective in minimizing egg losses.

2. Low production in January

- The farm should review environmental, feed, and flock management conditions in January to identify factors responsible for the reduced production and implement corrective measures for future cycles.

3. Higher contribution from Wing 2

- Management should investigate operational practices in Wing 2 and replicate effective strategies in Wing 1 to improve overall production balance.

4. High damaged eggs recorded by a specific staff member

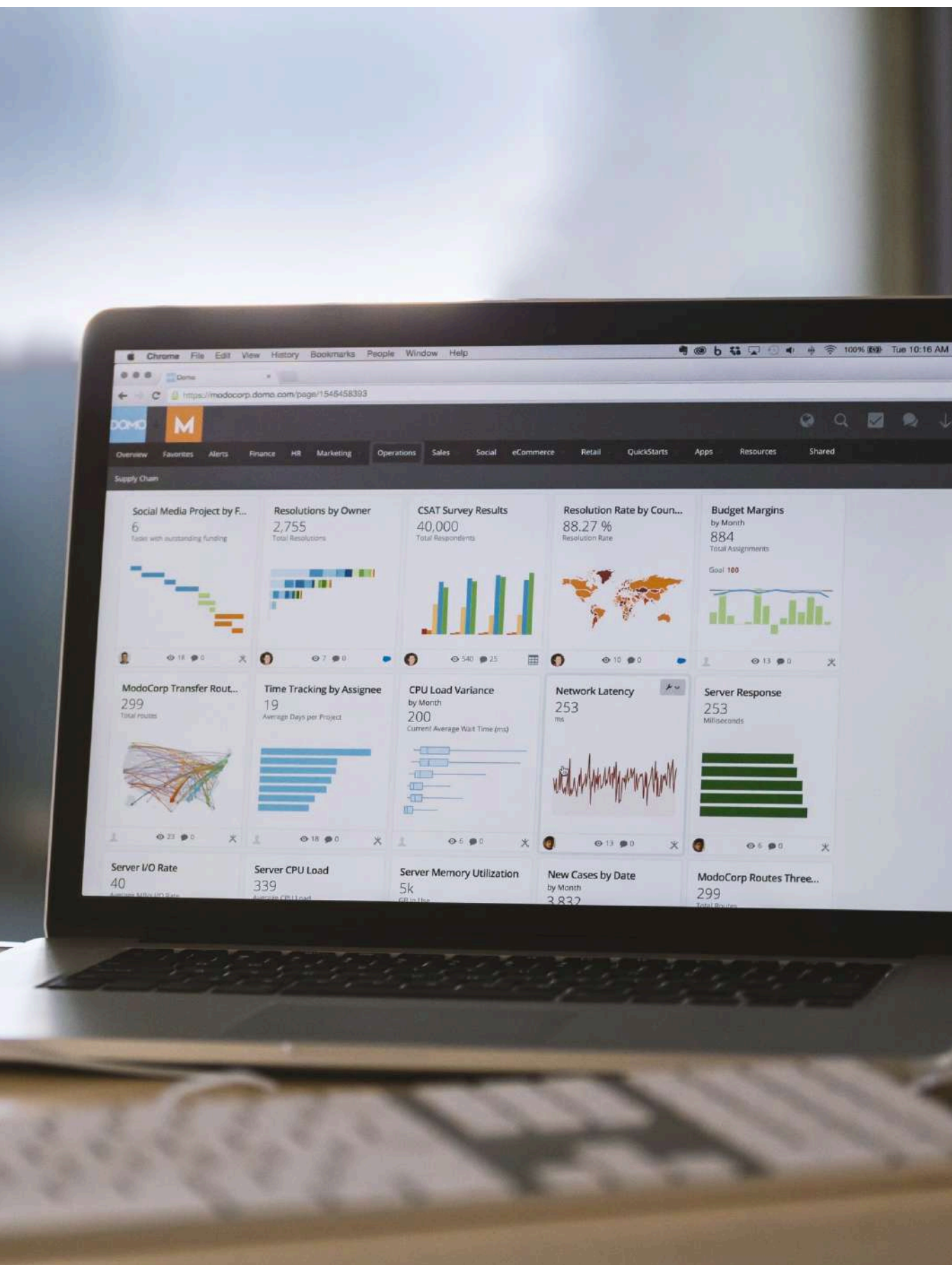
- Closer supervision should be provided to the affected staff, to reduce handling-related egg damage.

5. Lack of wing-level tracking for damaged and small eggs

- The data collection process should be updated to capture wing-specific records for small and damaged eggs to improve accountability and good analysis.



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CONCLUSION

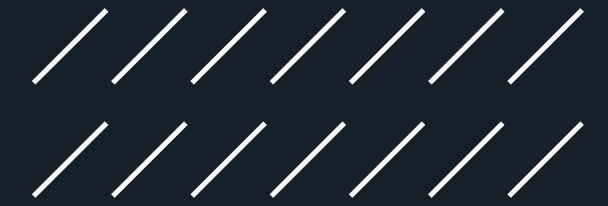
By combining a well-designed Google Form with a carefully scripted dataset restructuring process, Smile Planet Farms now has a robust, standardized, and future-ready data collection system.

This approach significantly improves data quality, reduces operational errors, and enhances the farm's capacity for data-driven decision-making.



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Thank You

Thank you for viewing this Data Analysis presentation.
Your attention is greatly appreciated.

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